VideoComposer: Compositional Video Synthesis with Motion Controllability

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Problem

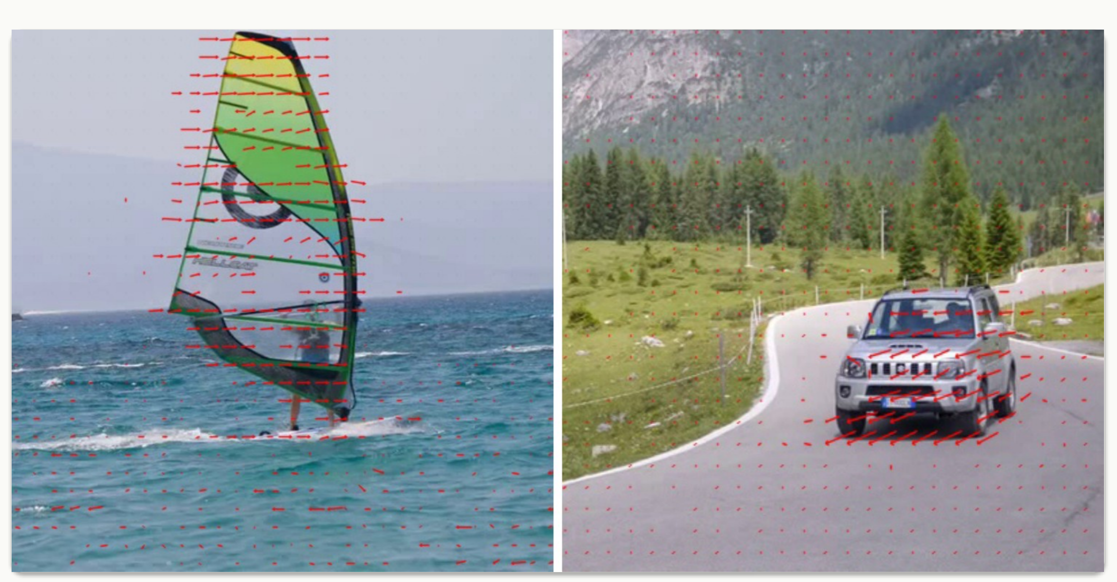
Controllable video synthesis is far harder than for images: temporal dynamics vary widely across clips, and every generated frame must stay temporally consistent.

> *Spatial controls designed for images give **no reliable handle** on how content moves and evolves over time.*

Motivation

- Use **motion vectors** from compressed video as an explicit temporal condition — direct guidance on dynamics.
  - Naively fusing heterogeneous conditions breaks consistency, motivating one unified space-time encoder.

To control a video, control its **motion** — not just its appearance.



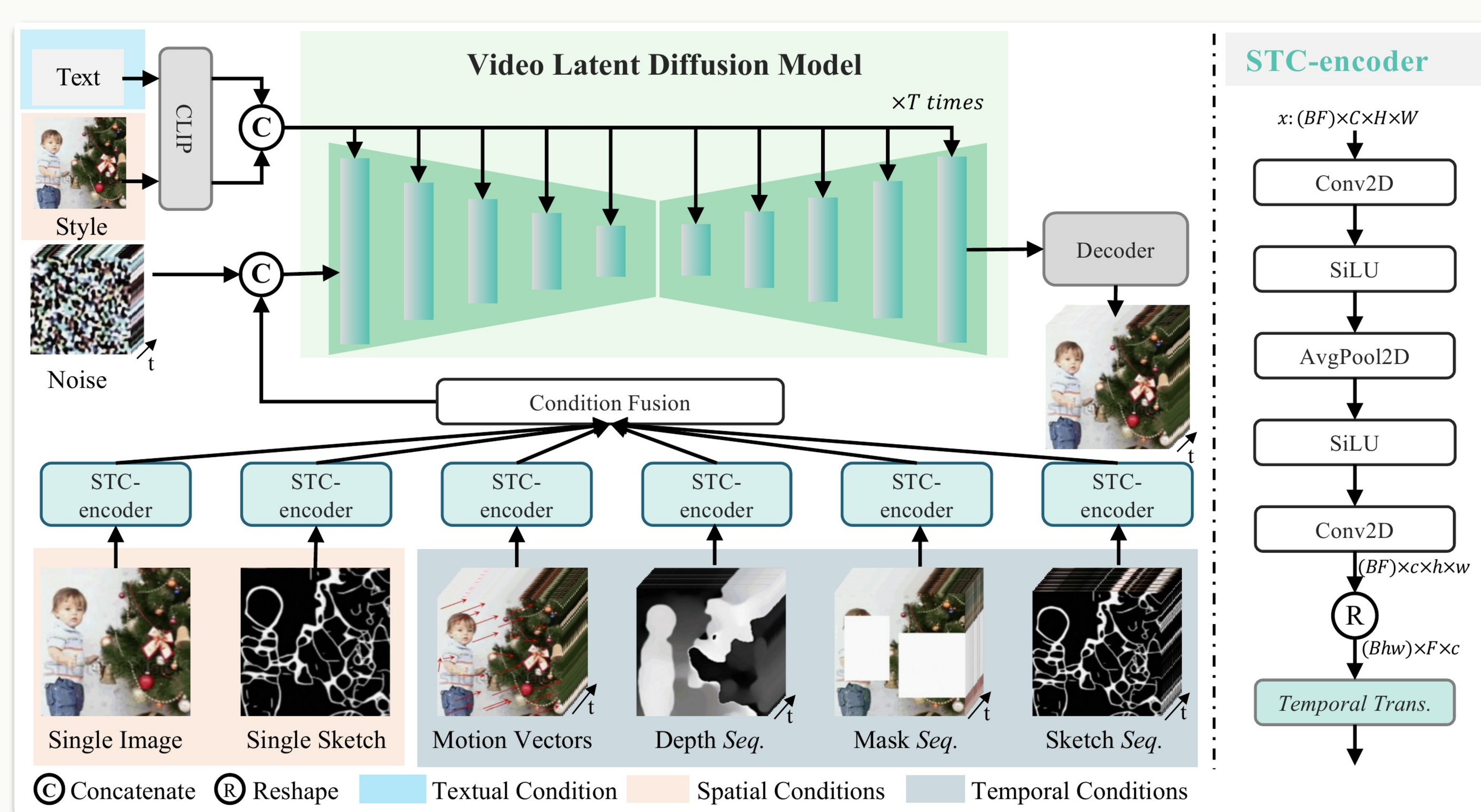
Motion vectors extracted from compressed video encode inter-frame variation, giving an explicit, cheap temporal-dynamics signal.

Method

- Decompose each video into **textual**, **spatial** and **temporal** conditions that can be freely composed.
- Sequential conditions pass through the **STC-encoder** (2× 2D conv → avg-pool → temporal Transformer), fused by element-wise addition.
- Fused controls are concatenated with the noisy latent; text & style enter via cross-attention. Two-stage training; DDIM + classifier-free guidance at inference.
- Motion vectors are extracted in standard compressed formats to encode inter-frame variation as a cheap, explicit temporal signal.

$$\mathcal{L}_{VLDM} = \mathbb{E}_{\mathcal{E}(x),\, \epsilon\sim\mathcal{N}(0,1),\, c,\, t}\left[\|\epsilon - \epsilon_\theta(z_t, c, t)\|_2^2\right]$$

denoise z_t under composed conditions c; ϵ_θ predicts the added noise



Overall architecture: a video is decomposed into textual, spatial and temporal conditions, embedded by the unified STC-encoder or CLIP, then jointly guide the VLDM denoiser.

Dataset / Benchmark

- Trained on **WebVid10M** (10.3M video–caption pairs) and the **LAION-400M** image corpus.
- Text-to-video quality measured on **MSR-VTT**; motion control on 1000 clips.

WebVid10M · LAION-400M · MSR-VTT

Key Results

Method (zero-shot)	FVD ↓	CLIPSIM ↑
Stage-1 T2V pre-train	803	—
VideoComposer	580	0.2932

−223 FVD from compositional training

SO WHAT → *Multi-condition control costs nothing in T2V quality — it even beats stage-1 pre-training.*

Ablation Study

- Adding **motion vectors** cuts motion-control error from **4.03 to 2.67** (lower is better).
- The **STC-encoder** lowers it further to **2.18** — both ingredients matter.

4.03	2.67	2.18
text only	+ motion vectors	+ STC-encoder

SO WHAT → *Motion vectors supply the temporal signal; the STC-encoder makes the model use it — sharpening motion control and inter-frame consistency.*

Headline Numbers

580
FVD · MSR-VTT (zero-shot)
vs 803 stage-1 pre-training

- **0.2932** CLIPSIM
- **2.18** motion-ctrl error
- **8+** condition types
- **10.3M** train pairs

Takeaway

By treating a video as a composition of textual, spatial and temporal conditions — and pairing compressed-video motion vectors with a unified STC-encoder — VideoComposer delivers flexible, controllable video synthesis with strong inter-frame consistency and no loss in text-to-video quality.

MOTION

One model driven by text, sketches, depth, masks, reference video, or two strokes — motion becomes an explicit, editable dimension.

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